

CHE 396 Quiz 2

Mariana St Peter

TOTAL POINTS

78 / 100

QUESTION 1

11 20 / 20

- ✓ - **0 pts** Correct
- **20 pts** wrong or not done
- **0 pts** Click here to replace this description.
- 💬 excellent!

QUESTION 2

22 15 / 20

- **0 pts** Correct
- **17 pts** only LK and HK identified.
- **20 pts** wrong or not done
- ✓ - **5 pts** HK calculation is wrong.

QUESTION 3

33 3 / 5

- **0 pts** Correct
- **5 pts** reflux ratio not mentioned.
- ✓ - **2 pts** reflux ratio stated as decreased.

QUESTION 4

44 0 / 5

- **0 pts** Correct
- ✓ - **5 pts** Incorrect

QUESTION 5

55 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** Incorrect

QUESTION 6

66 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** Incorrect

QUESTION 7

77 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** incorrect

QUESTION 8

88 5 / 5

- ✓ - **0 pts** Correct
- **5 pts** incorrect

QUESTION 9

99 0 / 5

- **0 pts** Correct
- ✓ - **5 pts** Incorrect

QUESTION 10

1010 10 / 10

- ✓ - **0 pts** Correct
- **10 pts** Incorrect

QUESTION 11

1111 10 / 15

- **0 pts** Correct
- **15 pts** Wrong or not done
- **10 pts** Not explained well
- **5 pts** Pressure not mentioned
- ✓ - **5 pts** C4s are not mentioned

CHE 396 SENIOR DESIGN Quiz 2

10/15/2019

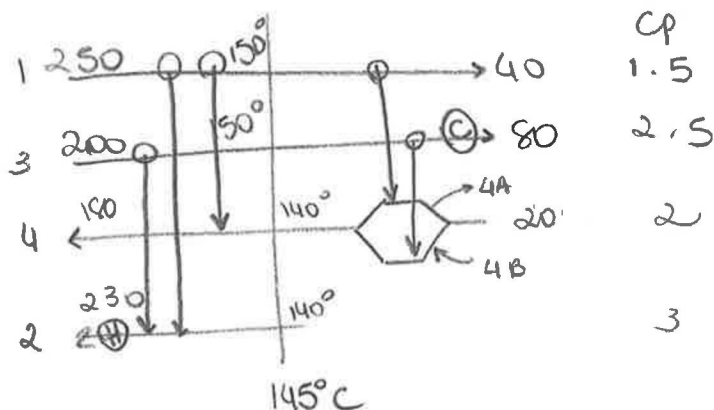
Mariana St. Peter

Name

- Determine the minimum utility requirements for the process set out below. Take the minimum approach temperature as 10°C . Design a heat exchanger network to achieve maximum energy recovery. Pinch point for the problem is 145°C .

Stream No.	Actual temperature ($^{\circ}\text{C}$)		Heat capacity flow
	Source	Target	rate CP ($\text{kW}/^{\circ}\text{C}$)
H 1	250	40	1.5
C 2	140	230	3
H 3	200	80	2.5
C 4	20	180	2

$\Delta T_{\min} = 10^{\circ}\text{C}$



below pinch [$ch \gg c_c$]

$$\Delta H_1 = 165 \text{ kW}$$

$$\Delta H_3 = 175 \text{ kW}$$

$$\Delta H_4 = 240 \text{ kW}$$

have to split ④

$$165 \text{ kW} = (140 - 20) CP_{4A}$$

$$CP_{4A} = 1.375$$

$$\therefore CP_{4B} = 0.625$$

above pinch [$ch \leq c_c$]

$$\Delta H_1 = 150 \text{ kW}$$

$$\Delta H_3 = 125 \text{ kW}$$

$$\Delta H_4 = 80 \text{ kW}$$

$$\Delta H_2 = 270 \text{ kW}$$

①—④ → transfer 80 kW

③—② → transfer 125 kW

①—② → transfer 70 kW

add a heater in stream ②

$$\text{heater} = 270 - 125 - 70$$

$$\text{heater} = 75 \text{ kW}$$

①—④A transfer 165 kW

③—④B transfer 75 kW

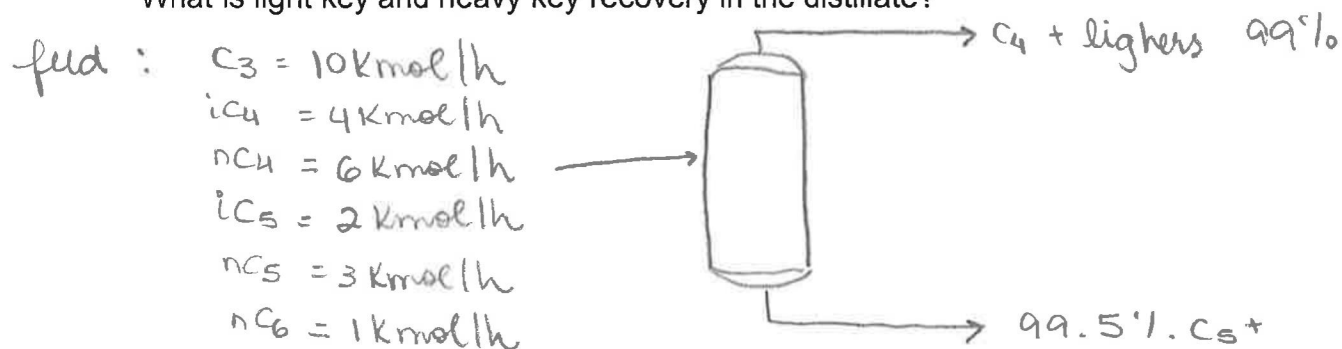
add a cooler in ③

$$\text{cooler} = 175 \text{ kW} - 75 \text{ kW}$$

$$\text{cooler} = 100 \text{ kW}$$

$$\therefore \text{total utility} = 175 \text{ kW}$$

2. The composition of the feed to a debutanizer is as follow; propane (C_3) is 10 kmol/h, isobutane ($i-C_4$) is 4 kmol/h, n-butane ($n-C_4$) is 6 kmol/h, isopentane ($i-C_5$) is 2 kmol/h, normal pentane ($n-C_5$) is 3 kmol/h and normal hexane ($n-C_6$) is 1 kmol/h. C_{5+} recovery in the bottoms is 99.5% and C_4 and lighter compounds recovery in the distillate is 99%. What is light key and heavy key recovery in the distillate?



light key nC_4

$$C_3 + iC_4 + nC_4 = 20 \text{ kmol/h} \times 99\% = 19.8 \text{ kmol/h}$$

which will be 100% recovery of $C_3 = 10 \text{ kmol/h}$

100% recovery of $iC_4 = 4 \text{ kmol/h}$

and only $(19.8 - 10 - 4) = 5.8 \text{ kmol/h}$ of nC_4 .

heavy key = iC_5

in the distillate will be 0.5% of 2 kmol/h

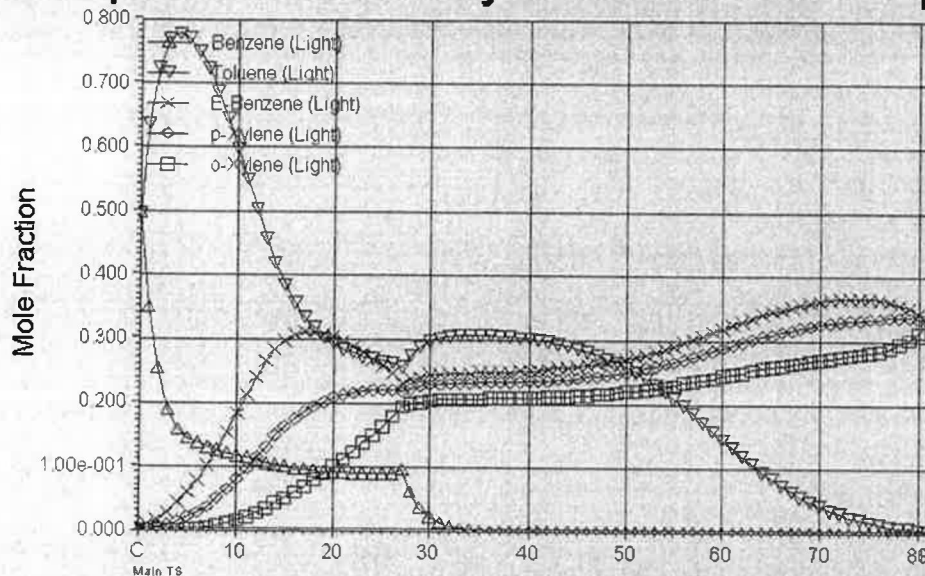
$\therefore 0.01 \text{ kmol/h}$ of iC_5 will be in the distillate.

distillate total composition;

$$\begin{aligned} C_3 &= 10 \text{ kmol/h} \\ iC_4 &= 4 \text{ kmol/h} \\ nC_4 &= 5.8 \text{ kmol/h} \\ iC_5 &= 0.01 \text{ kmol/h} \end{aligned}$$

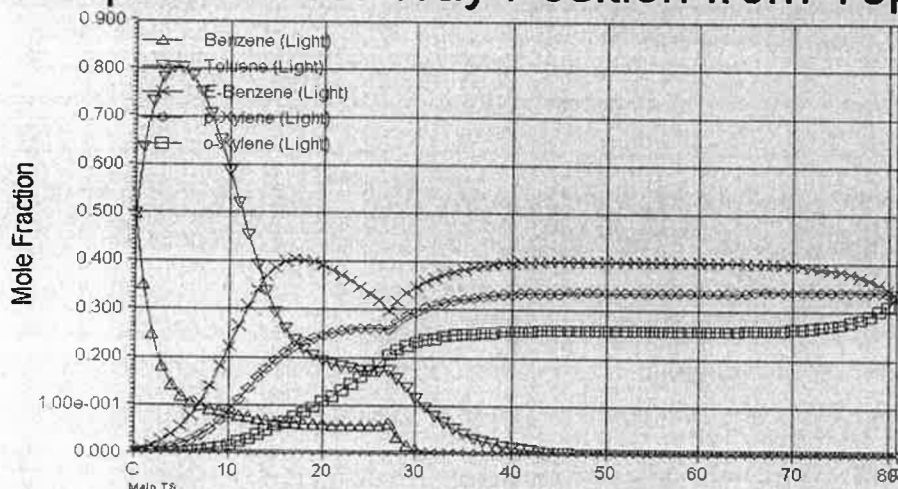
3. Equimolar mixture of benzene, toluene, ethylbenzene (EB), orthoxylene (OX) and paraxylene (PX) is separated by distillation. Composition vs tray position was graphed as:

Composition vs. Tray Position from Top



Later some parameters in the rigorous column has been changed and graph below is generated. Based on the graph below, which parameter has been changed, decreased or increased?

Composition vs. Tray Position from Top



reflux ratio has changed, it has decreased,

4. Distillation processes can only separate two species ("A" and "B") if:
- A. They have similar molecular weights.
 - ☒ B. They have very different enthalpies of vaporization
 - C. They have different vapor pressures.
 - D. One species exists as a vapor (at STP), while the other species must be a liquid.
5. The pressure in a tray column
- A. Is higher at the top.
 - ☒ B. Is higher at the bottom.
 - C. Depends on the type of tray and the tray design.
 - D. b and c.
 - E. None of the above.
6. The temperature in a tray column
- A. Is higher at the top.
 - ☒ B. Is higher at the bottom.
 - C. Is the same on every tray.
 - D. Could be a or b, depending on the components involved.
 - E. None of the above.
7. If a column design results in one that is too tall, which of the following is a possible remedy?
- A. Decrease the reflux ratio.
 - B. Reduce the number of trays.
 - C. Decrease the tray spacing.
 - ☒ D. b and c.
 - E. All of the above.
8. Which of the following is true about the feed to a distillation column?
- A. The column feed determines the column pressure.
 - ☒ B. The column feed must be larger than the column pressure.
 - C. The column feed tray is always in the middle of the column.
 - D. b and c.
 - E. None of the above.
9. Which of the following is true about the feed location in a distillation column?
- A. Feed on the middle tray.
 - ☒ B. The feed should be at the tray with approximately the same temperature.
 - C. The feed should be at the tray with approximately the same composition.
 - D. b and c.
 - E. All of the above.

10. What was the pressure of MTBE column in your mini-project 2 design?

20 bar

11. Methanol boiling point is higher than MTBE boiling point at 1 atm. And they form azeotrope. However, in your mini-project problem 2, MTBE is separated from methanol by distillation and MTBE is the bottom product. Explain how that column works.

We had to find the pressure where MTBE would have a higher boiling point than methanol, so methanol could be the light key in the distillate, while MTBE was the heavy key.

